

Software Engineering

How To Create Software Products Everyone Loves!

Module 6

UI/UX Fundamentals: Introduction

Ali Samanipour

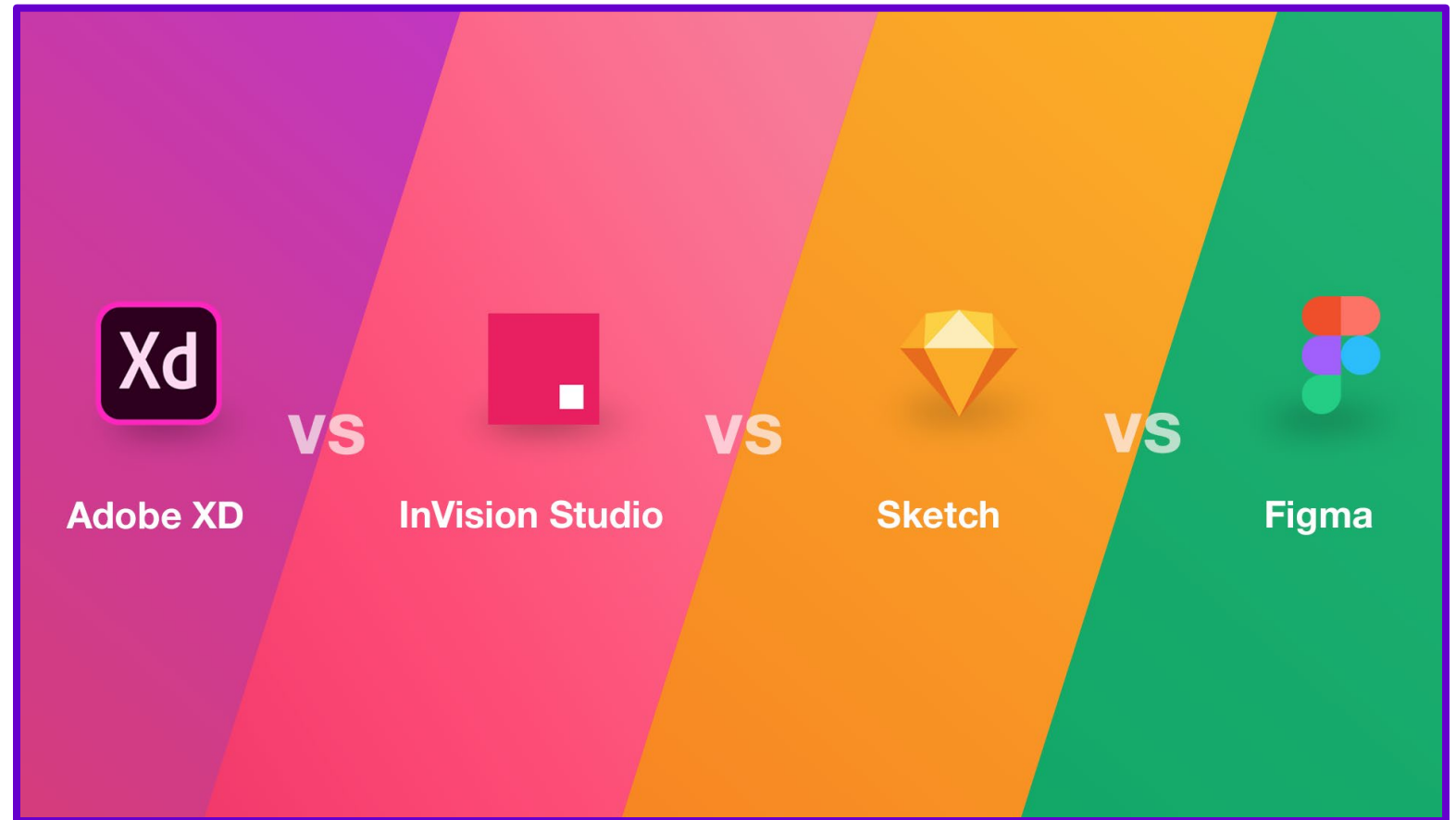
May. 2023

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Which Software Is Right For Me?

These are
just tools,
don't stick to
them

Stick with
Concepts and
Fundamentals



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Who is this course for?

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UI vs UX vs CX

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Design and Art

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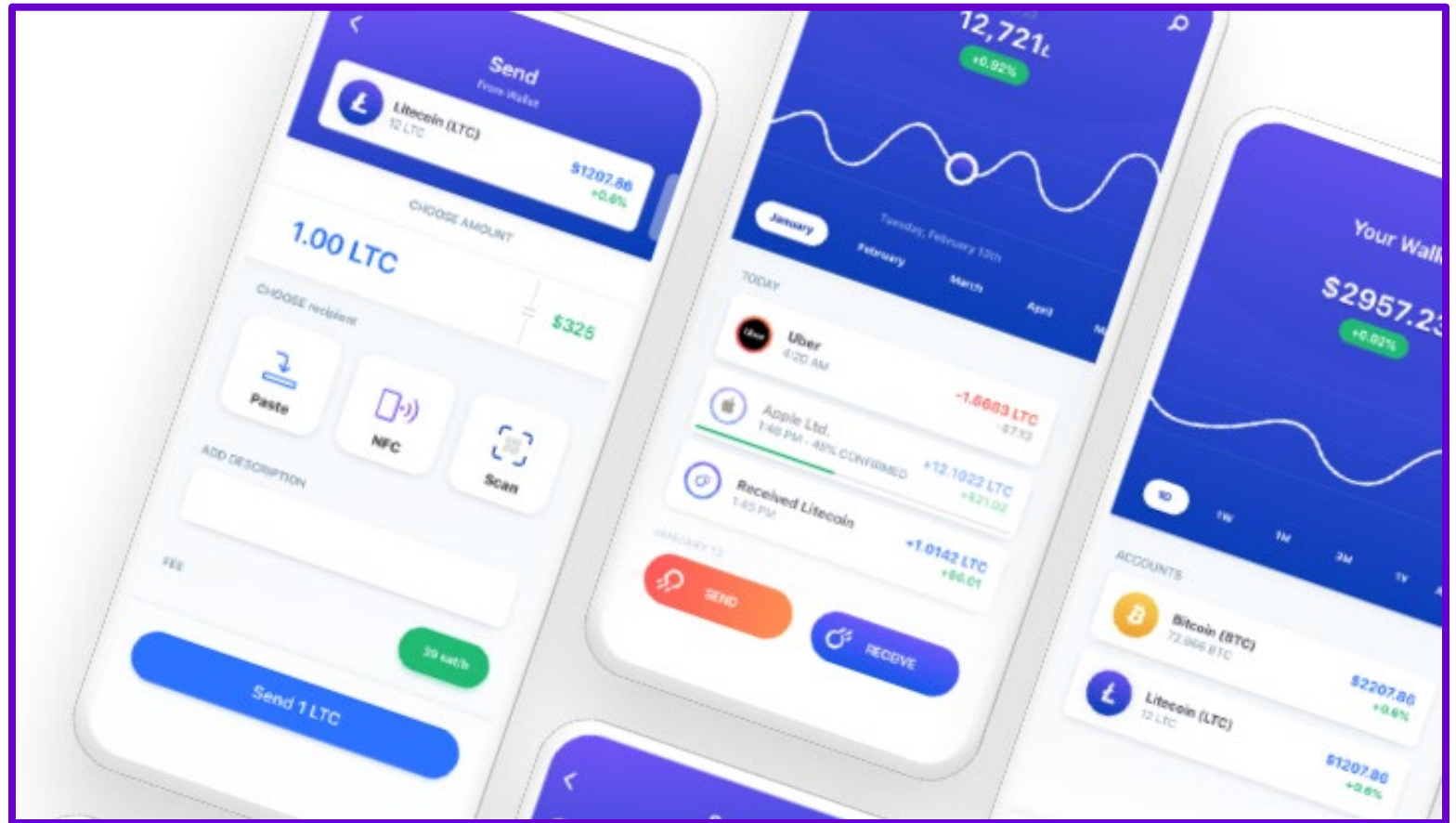
What makes UI good?

6

Perception

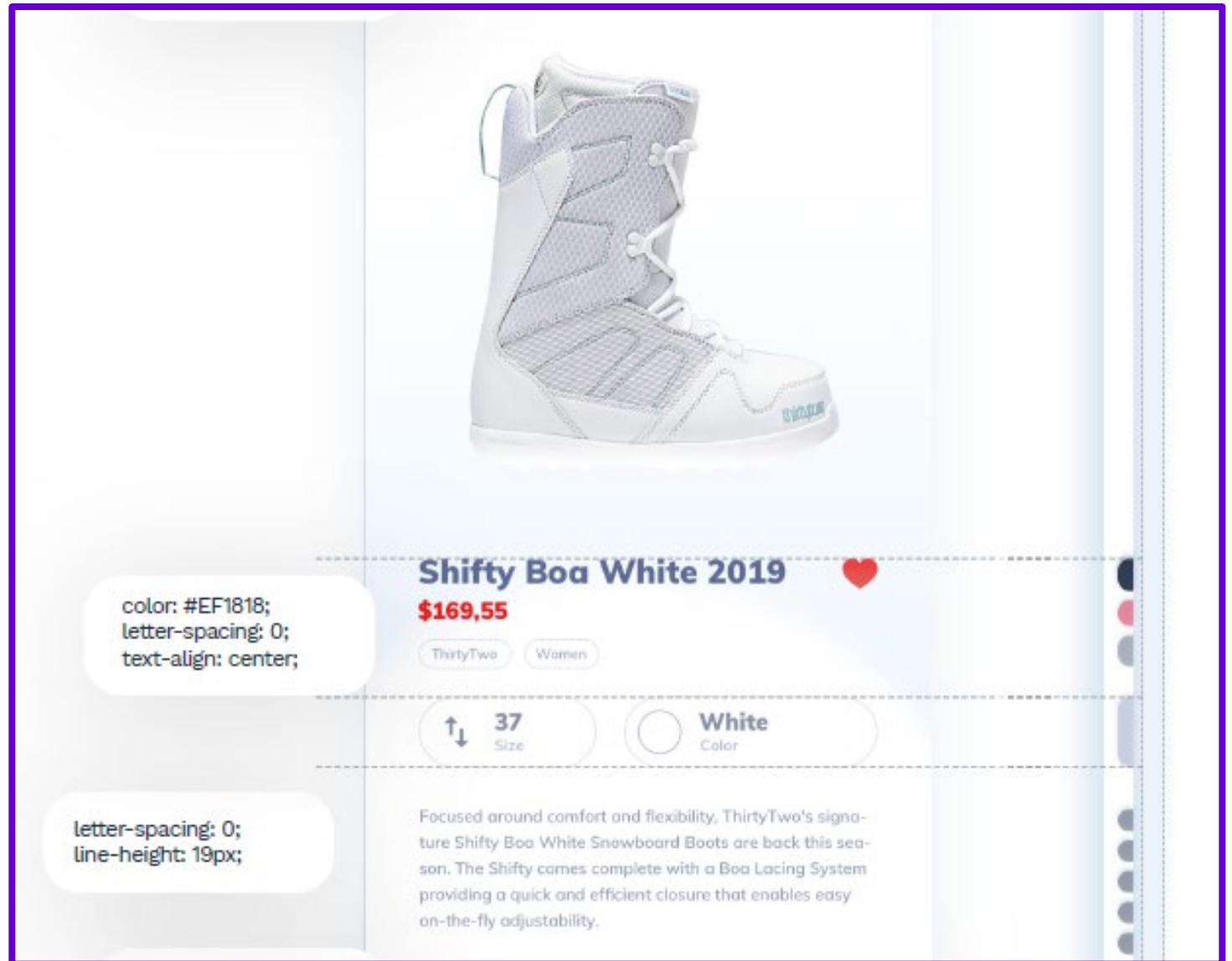
Who is this course for? (Designers)

Designers will gain knowledge on how to make better-looking and more usable products.



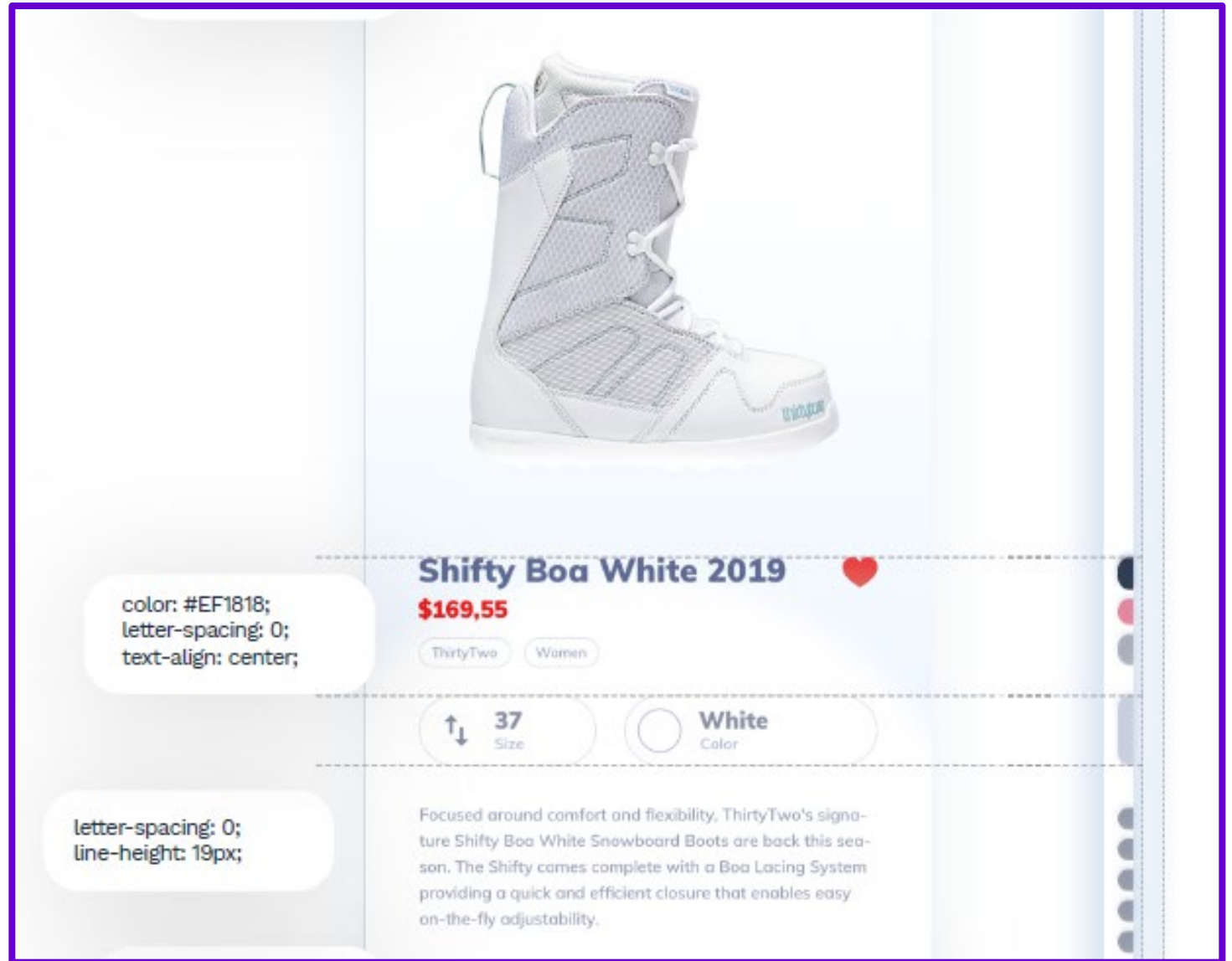
Who is this course for?
(Front-end
Developers)

will learn the basics
of design and best
practices for working
with designers to
speed up and raise
the quality of their
Implementation
work.



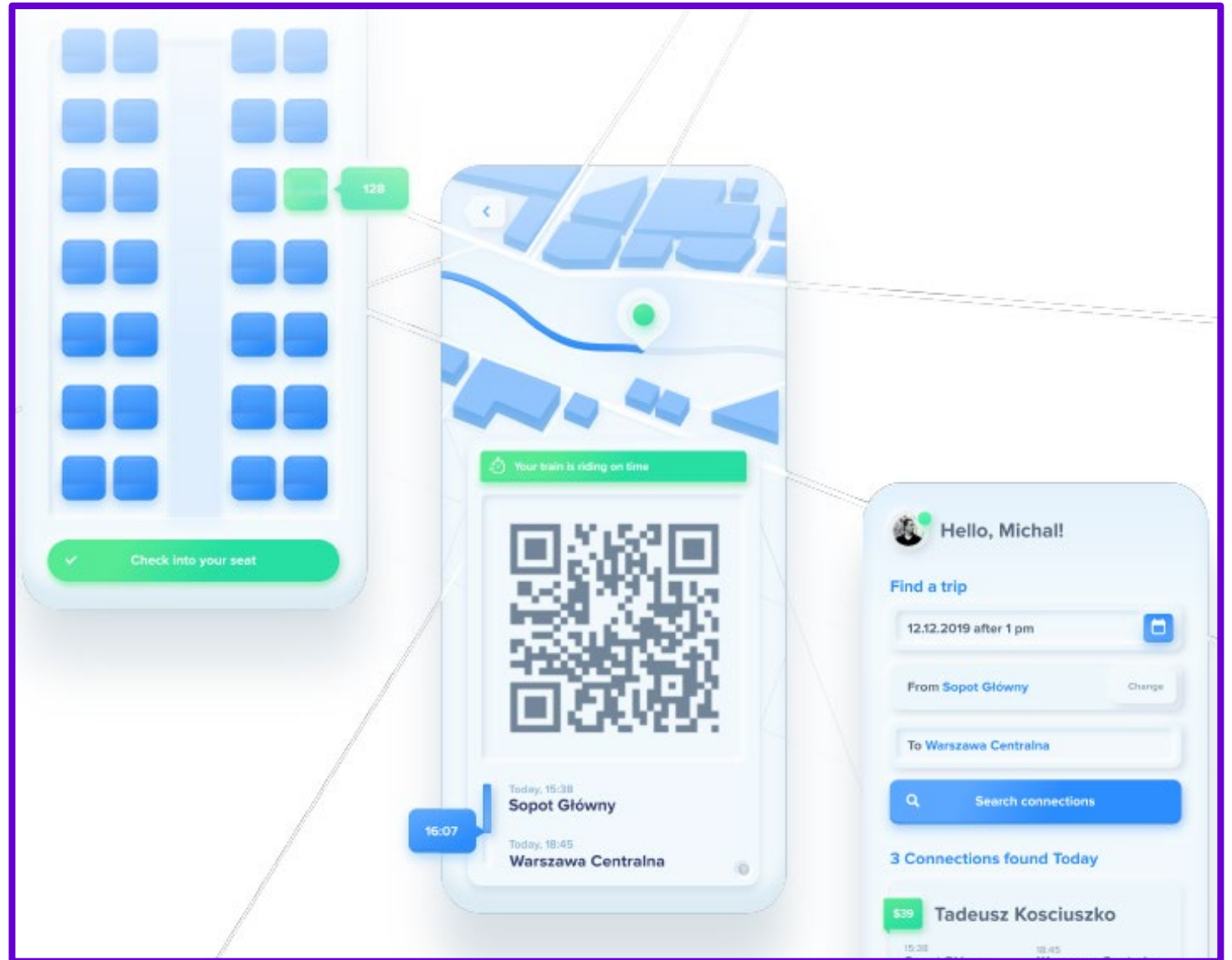
Who is this course for?
(Front-end
Developers) ...

Every designer
should understand
a bit of coding, and
every developer
should know what
design is all about



Who is this course for?
(Startup founders)

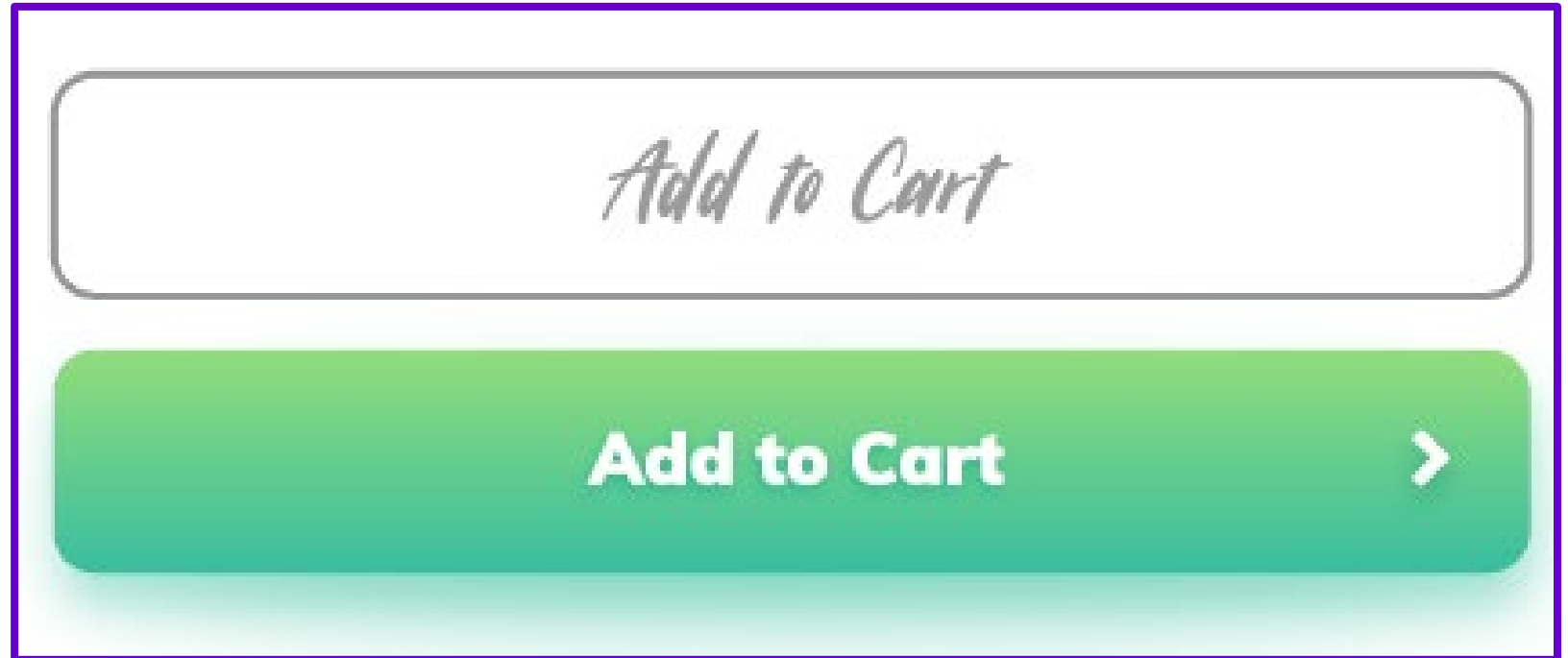
People "buy" into a
product or service
with their eyes first.



Who is this course for?

(Marketing and social media ninjas)

We all want
happy,
engaged
users
interacting
with what we
do.



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A software production line!

(Software Development Life Cycle-SDLC)

1.
Plan



Product
owner

Product
manager



2.
Analysis



Project
manager

Software
architecture



3.
Design



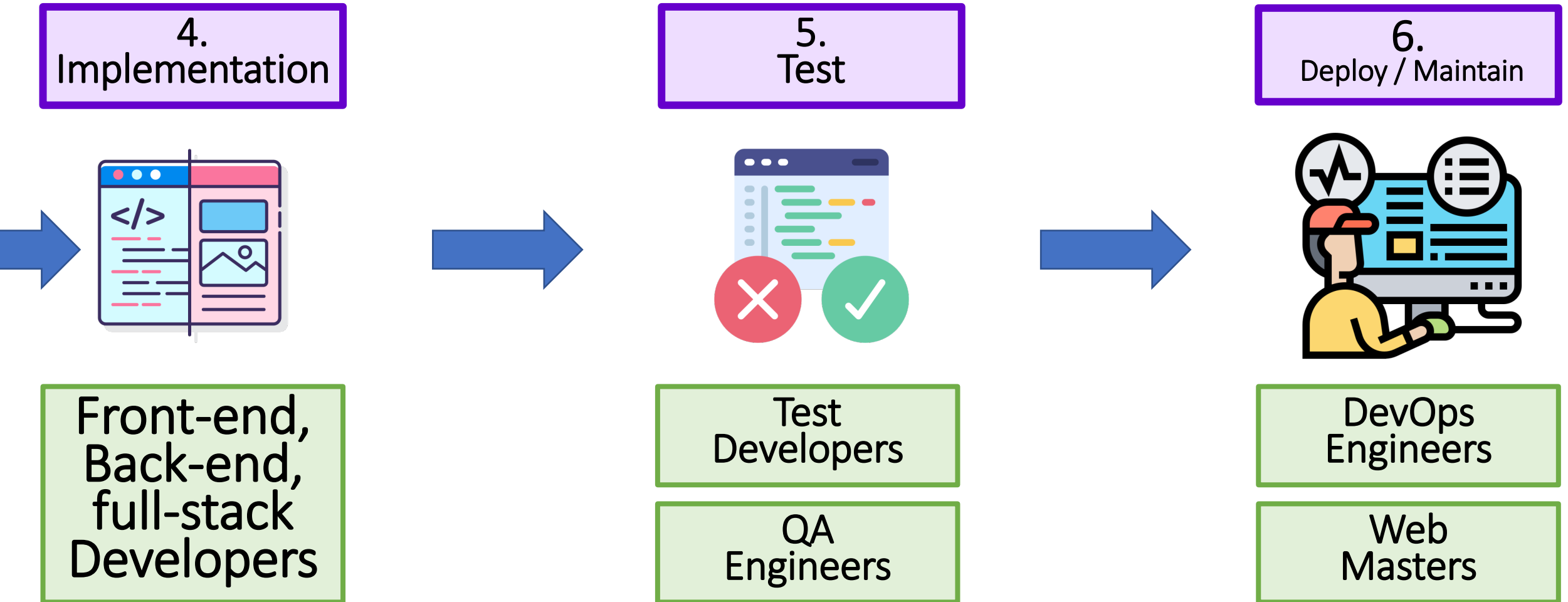
UI/UX
Designer

Technical
Designer



A software production line!

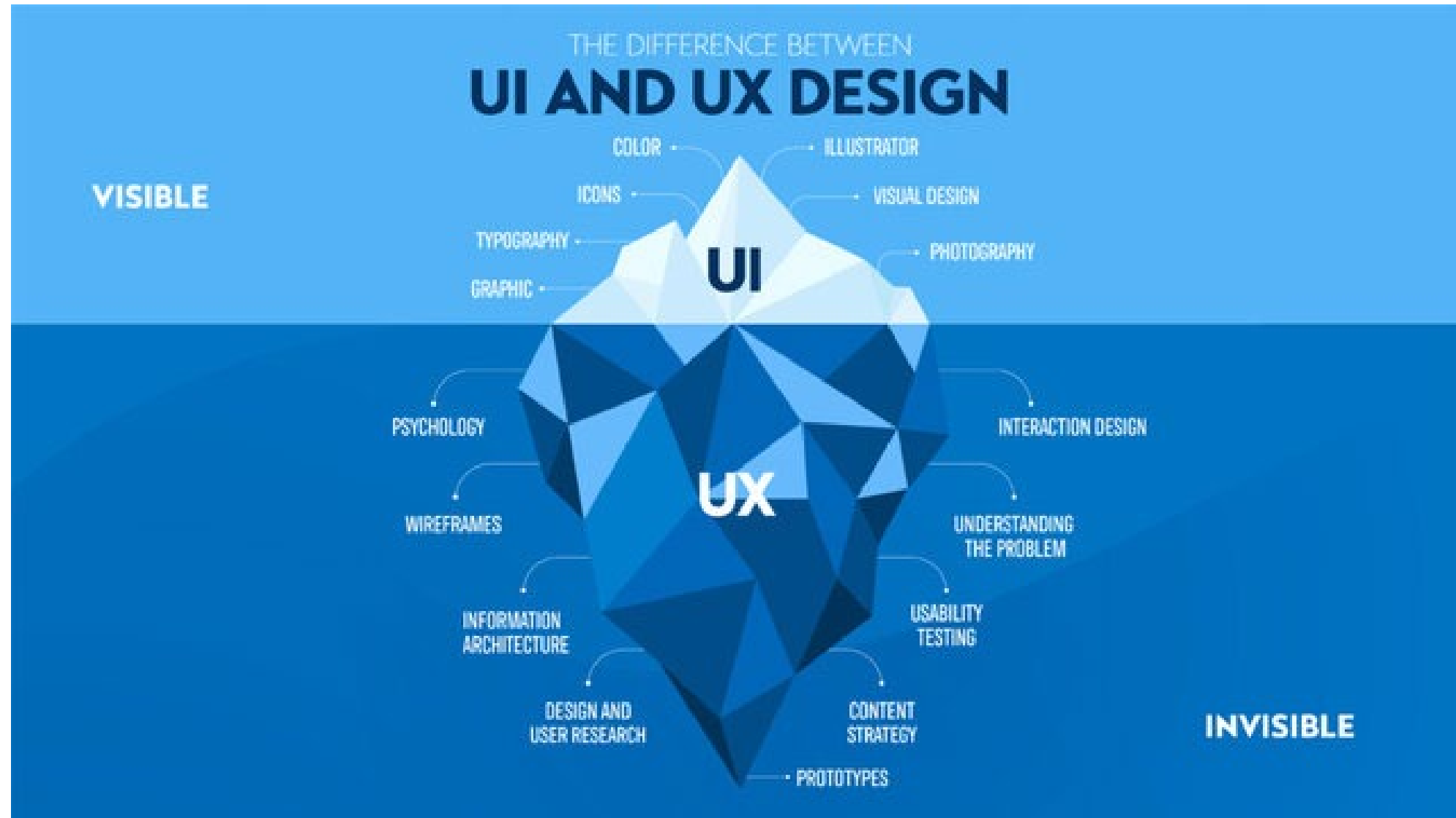
(Software Development Life Cycle-SDLC)



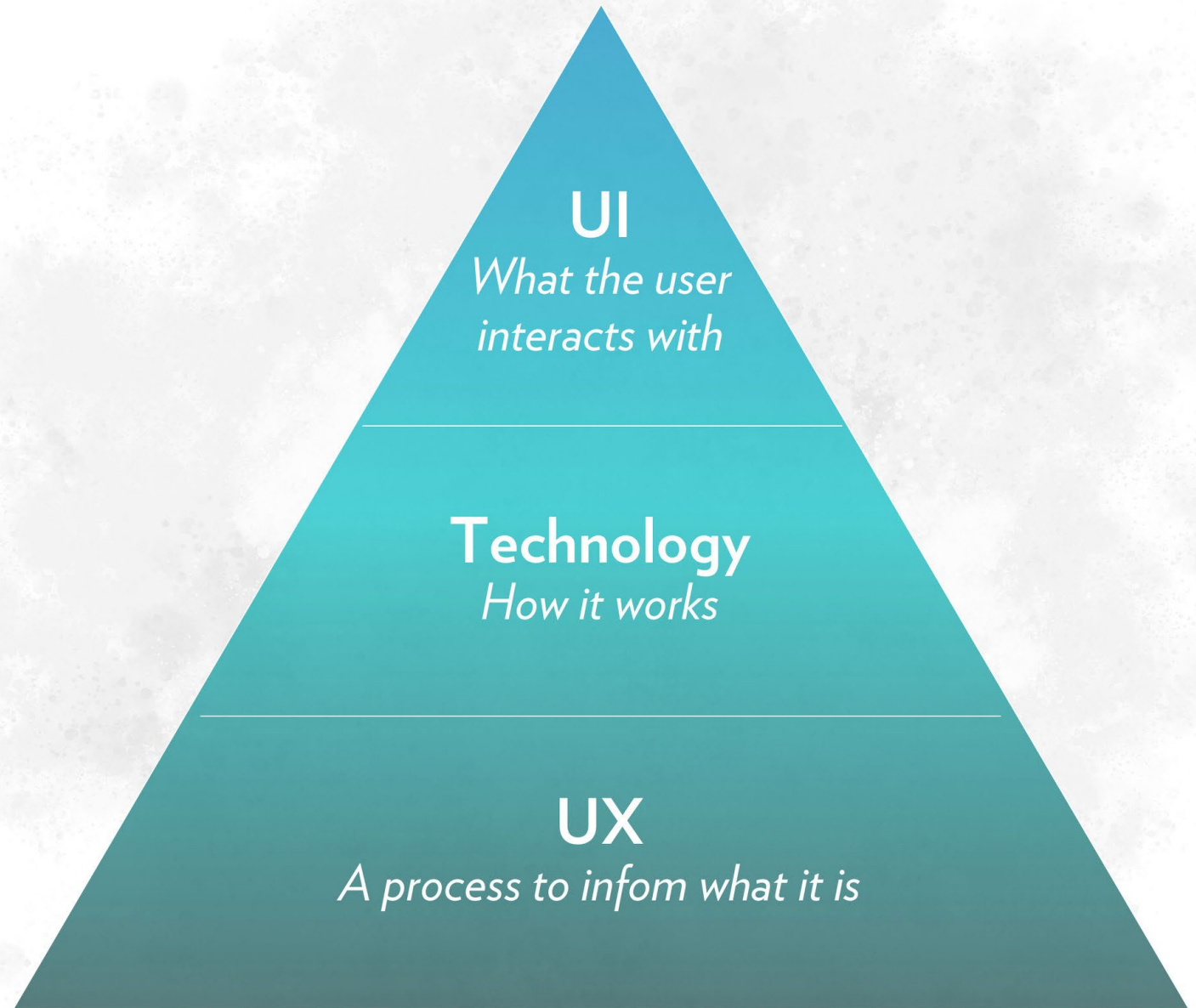
UI vs UX



UI vs UX

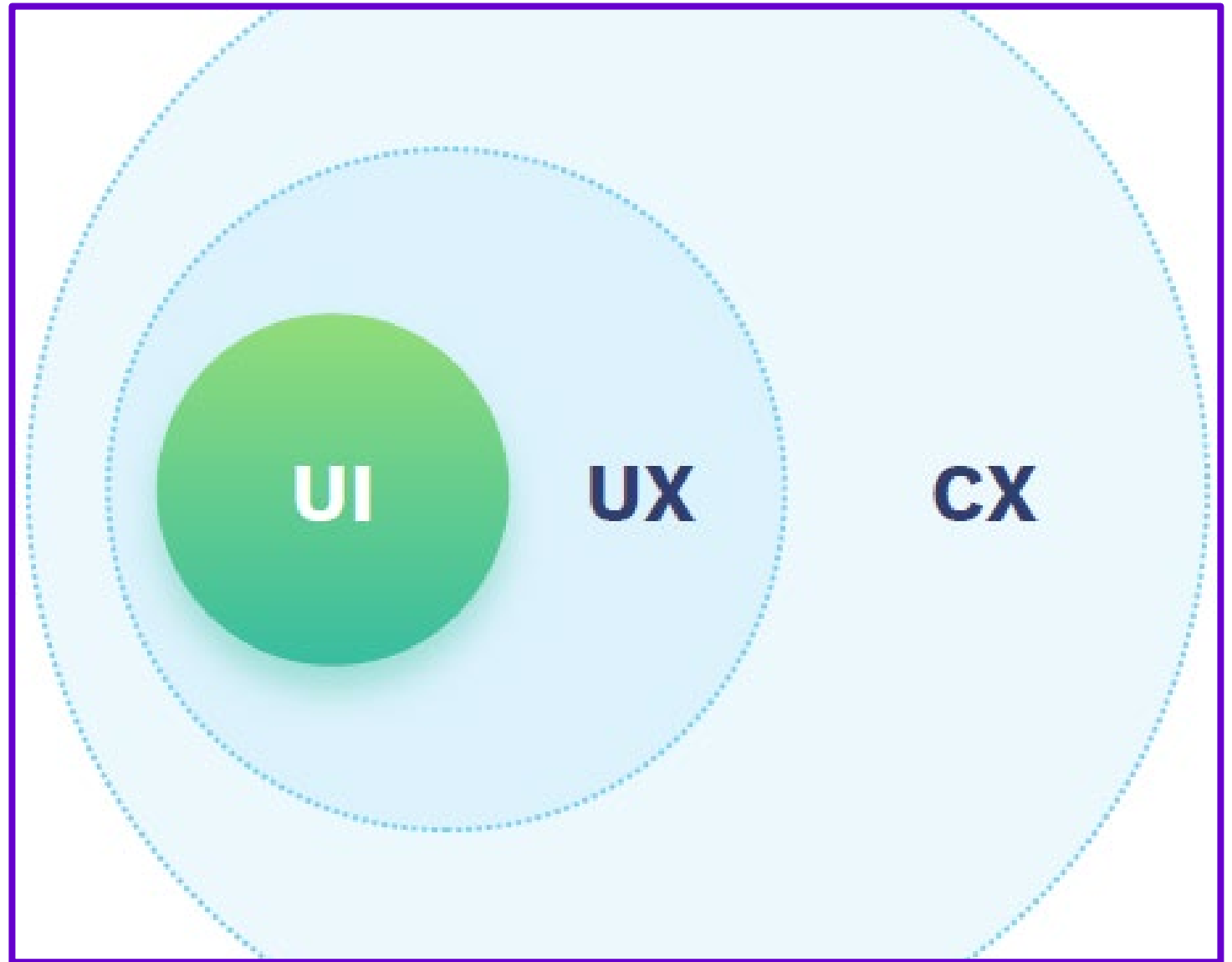


UI vs UX



UI vs UX vs CX

Customer experience is a high-level category with UX being a part of it, while UI being a part of the UX process



UI or User Interface

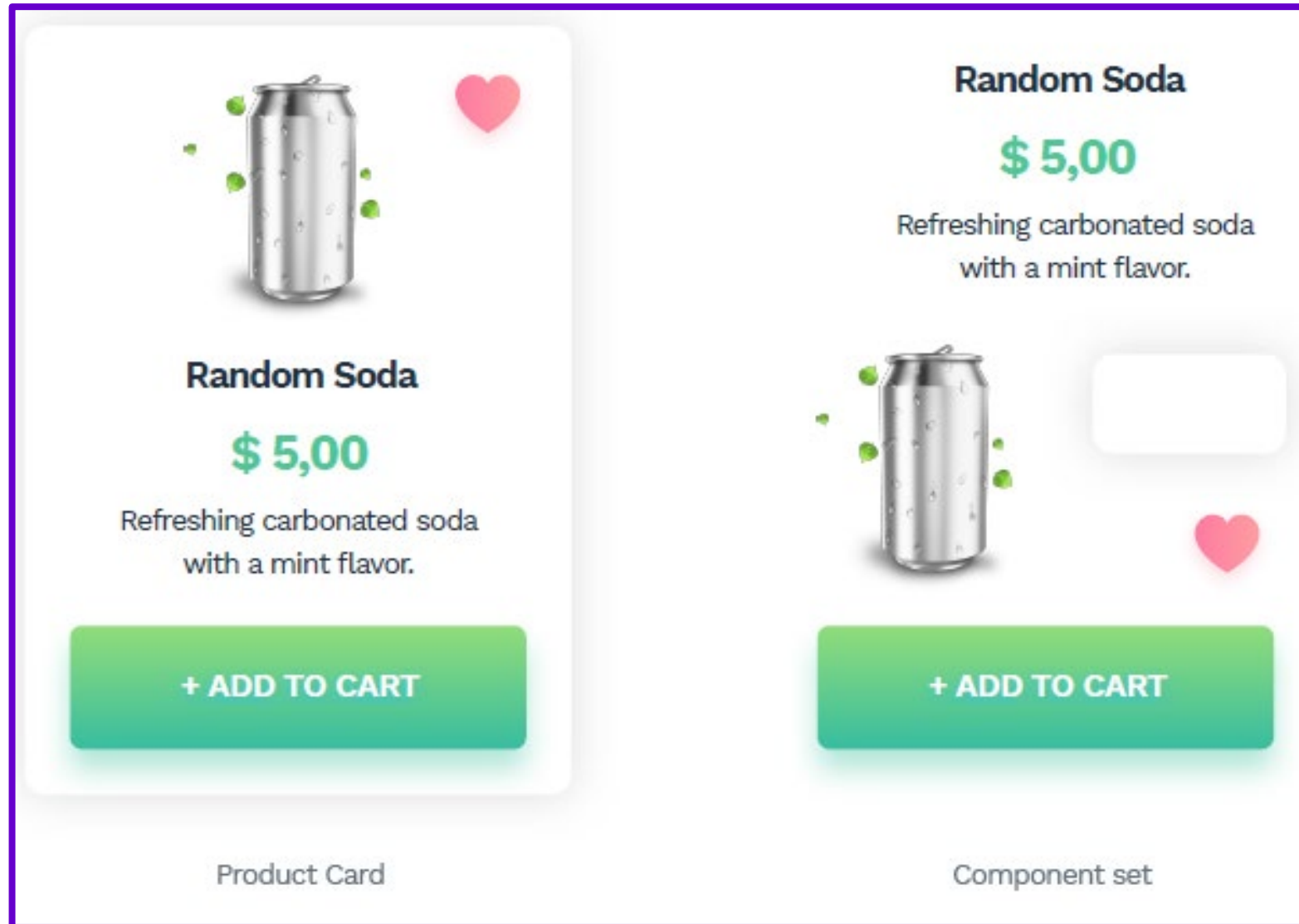
UI is the visual representation of a digital product. Its obvious use cases are apps and websites

UI is the link between the user and the functionality of a product.

It helps to achieve the desired outcome through a series of human-machine interactions.

UI or User Interface

It's a series of text, shapes, graphics, and photographs combined in a way that allows for a fluid, natural interaction.



UI or User Interface

An interface is a mix of the grid, layout, typography, colors, animations, and micro-interactions. In other words, UI is all that we experience – mostly with our eyes.

UX, defined as information architecture is a blueprint for UI's refined final look

UI Designer

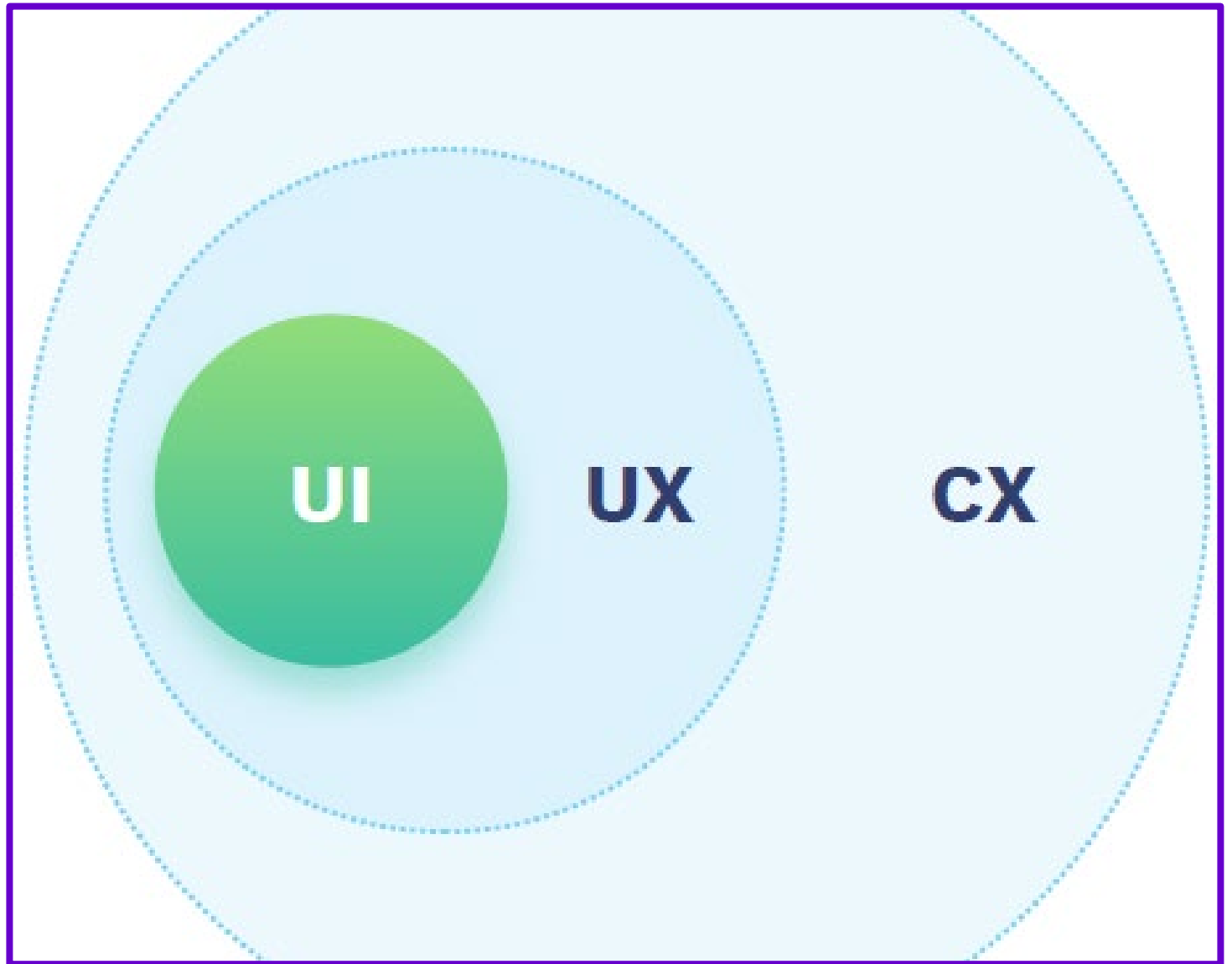
A UI designer is responsible for the final look & feel of the product.

The job is to define its unique style and make it fit the target market

That visual representation should be readable, usable, and devoid of any unnecessary, distracting elements

Good UI

“An interface is like a joke. If you have to explain it, it’s not that good”



UX or User Experience

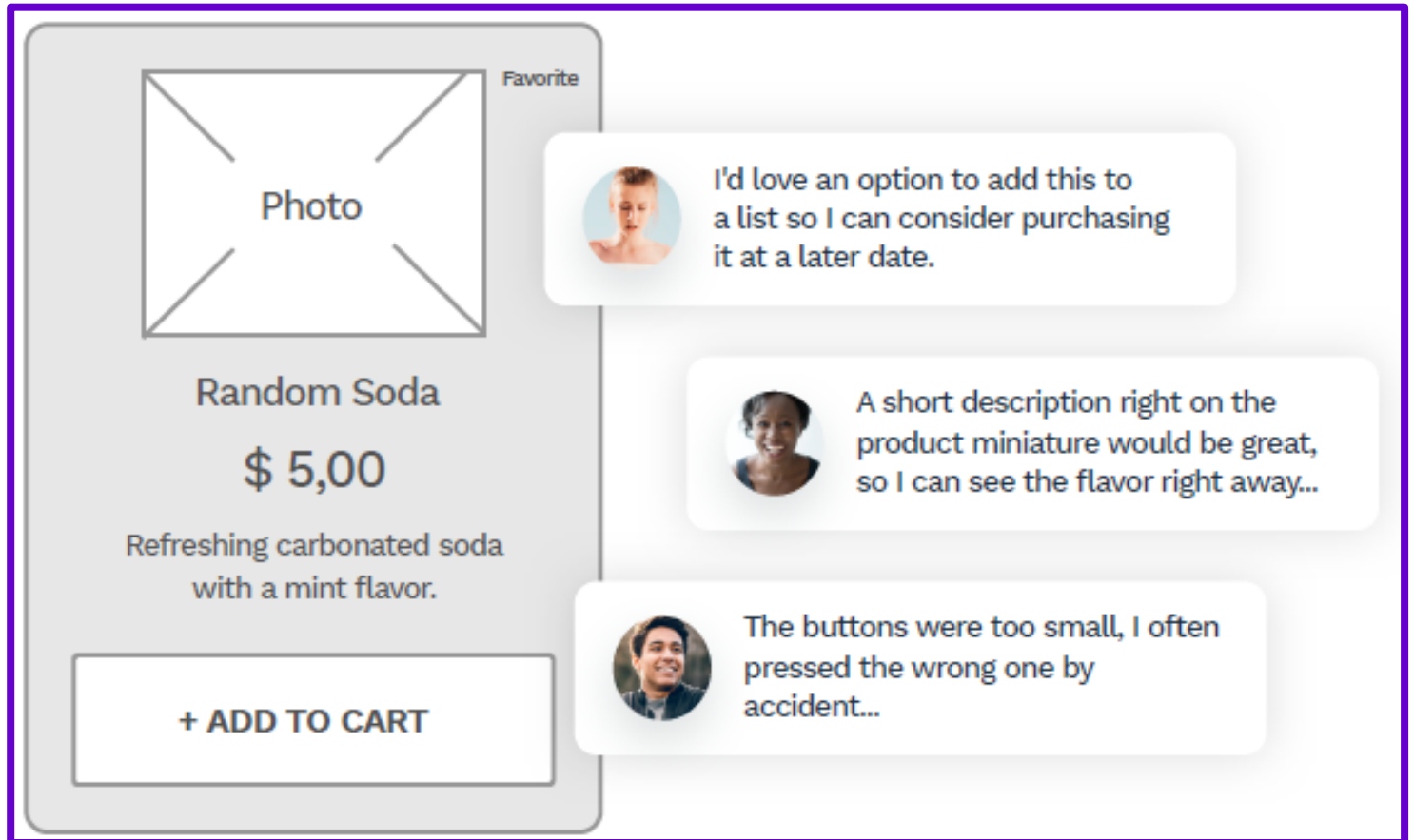
UX defines and studies how easy it is to use the product. With digital products, that means the interface, navigation patterns, and communications

The goal of UX is to allow the largest possible group of users the ability to understand and use a product.

If the role of a UI designer is the look & feel, the role of a UX designer is defining how it will work.

UX or User Experience

Aside from information architecture planning, UX also covers various kinds of research (surveys, A/B tests, focus groups, interviews, workshops, ...).



CX or Customer Experience

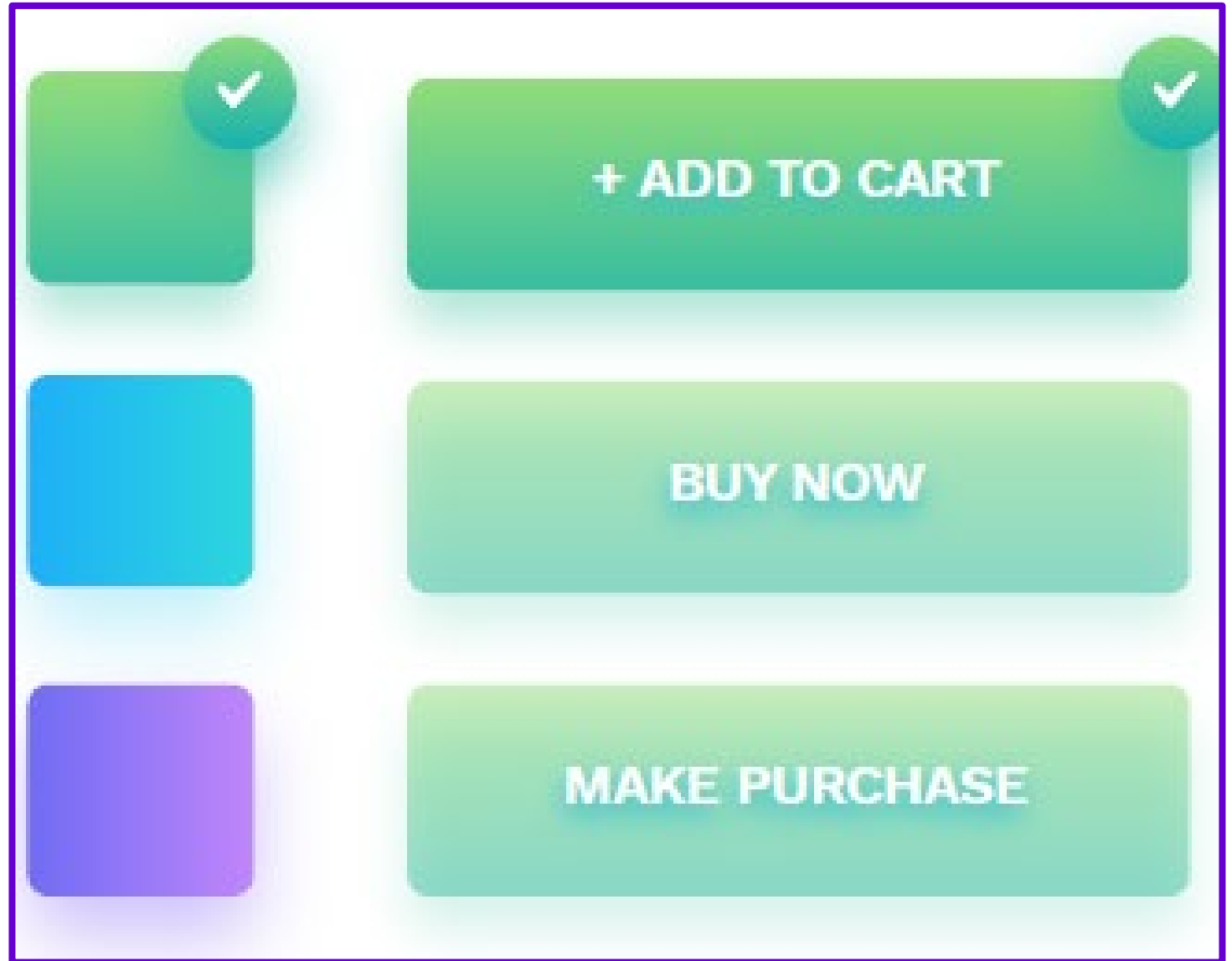
CX is a top-level process that defines not only how your product works, but how your entire company operates

The CX process is the UX of your product and your branding, marketing, or friendly customer support on the hotline.

You can design Customer Experience by defining clear, consistent rules for your entire brand. Those rules apply to all real-world experiences, as well as UX and UI.

CX or Customer Experience

All of those elements combine for a consistent, coherent vision with which our potential customer can quickly identify



Product Designers

A product designer can help with both the business processes, choosing the right building approach, and final interface designs.



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Design and Art

A graphic designer is commonly associated with being an artist, while UI design is more of a craft, that adheres to a set of strict rules.

User interfaces are not "painted with pixels," as many people think about them. They're a precise, visual blueprint for a working product.

They're not artworks on a gallery wall, but rather an architectural plan - accurate and with very little artistic chaos.

Design and Art

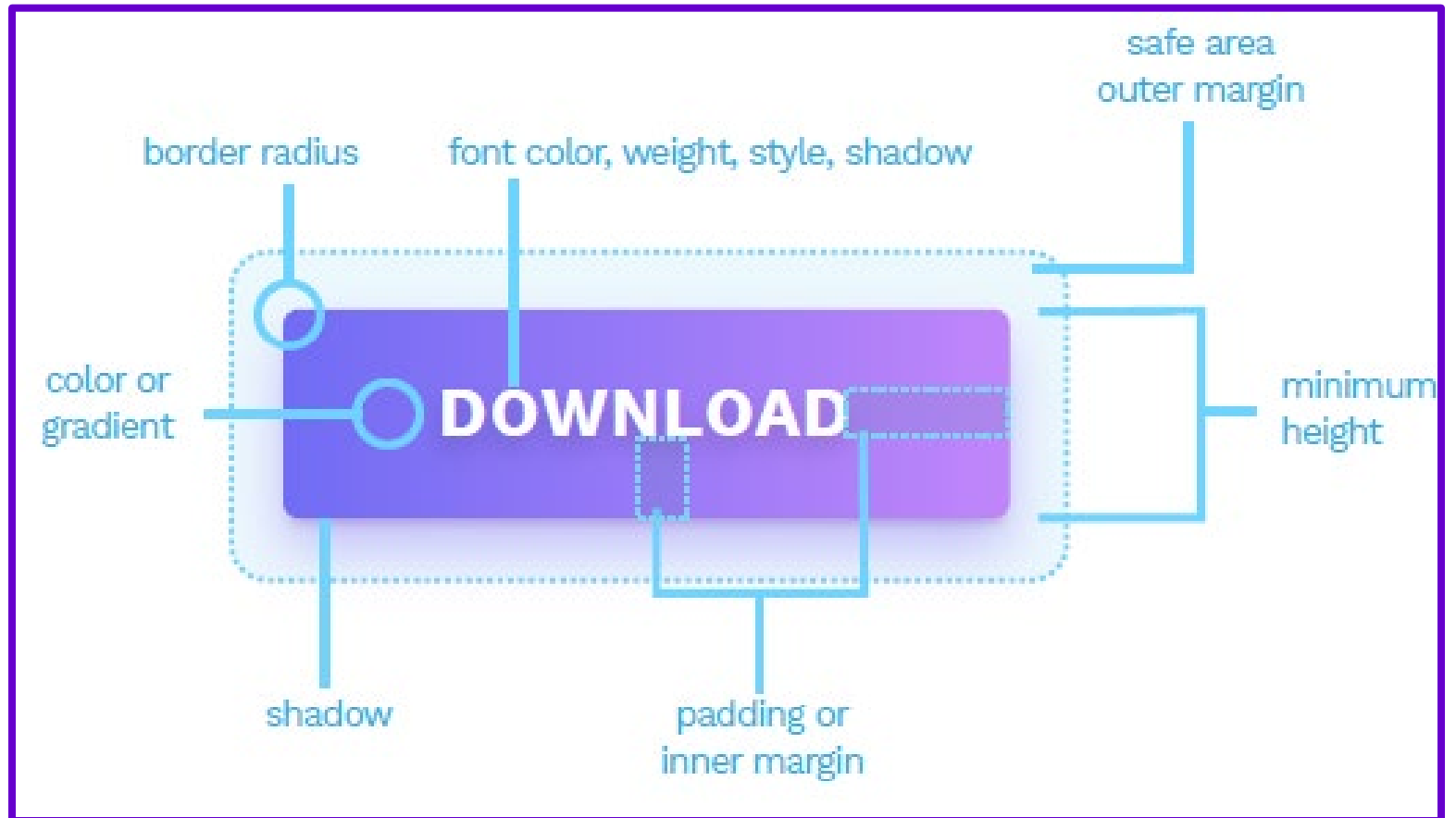
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UI Design and Art

Great interface design is pretty rigid in terms of its rules, constraints, and regulations.



UI Design is NOT art

Readable,
useful, and
aesthetically
pleasing
result is the
culmination
of a
successful UI
design



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Good UI?

Great UI is a sum of many parts. It is the interplay of colors, typography, interactions, layout, and style that requires a lot of work, experience, and imagination.

If you're starting as a junior designer, try to use the tested good practices first. Innovation and your very own style will come naturally, with experience.

10 Rules of Good Design by Dieter Rams, Good Design ...

IS INNOVATIVE

MAKES A PRODUCT USEFUL

IS AESTHETIC

IS LONG-LASTING

MAKES PRODUCT UNDERSTANDABLE

IS UNOBTRUSIVE

IS HONEST

THOROUGH DOWN TO THE LAST
DETAIL

IS ENVIRONMENTALLY FRIENDLY

AS LITTLE DESIGN AS POSSIBLE

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Visual Perception

Our brain is always trying to fill the gaps.

The brain subconsciously tries to connect the dots and see an image as a whole, instead of a subset of smaller elements

We group and connect similar objects and easily differentiate things that don't belong in the group

Perceptive visual mechanism

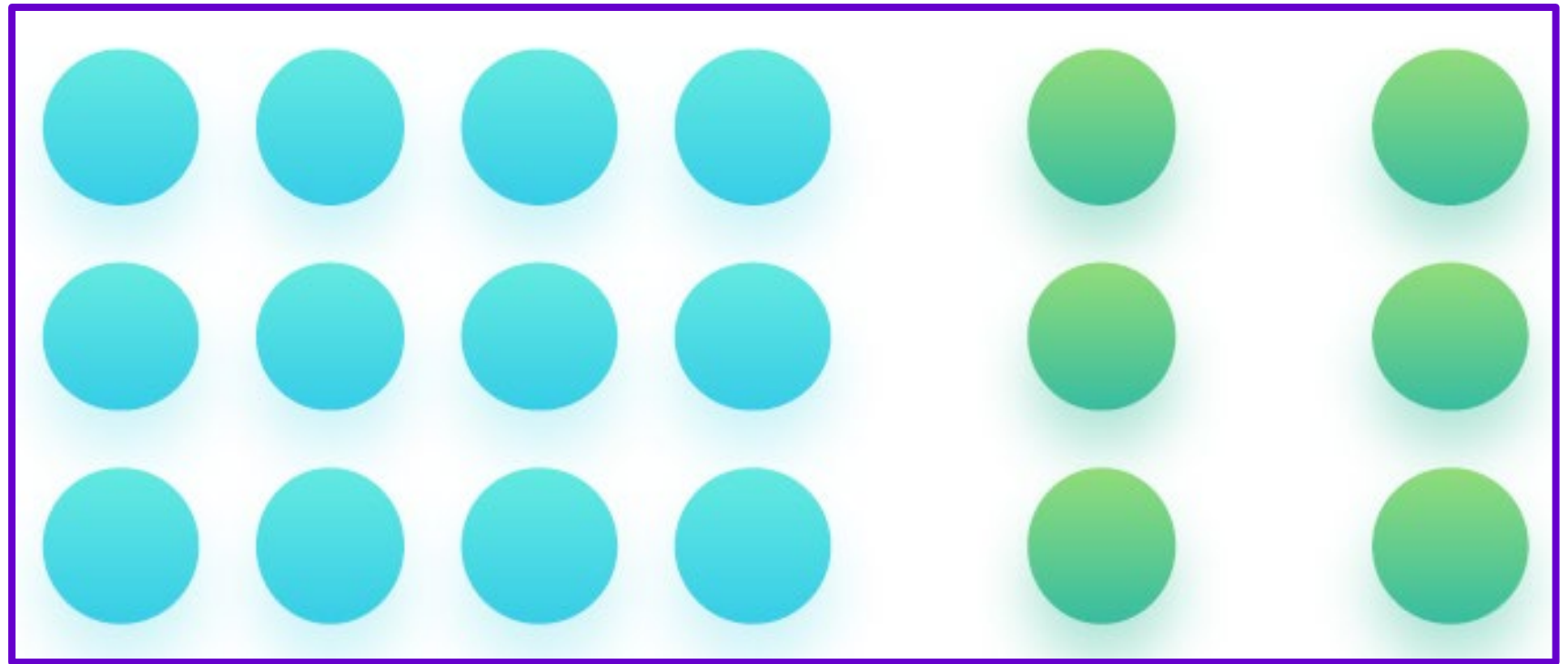
Rules of perception are perceptive mechanisms our mind uses to connect and organize partial information into a coherent whole.

Knowing and understanding these methods is a crucial element to a successful user interface design

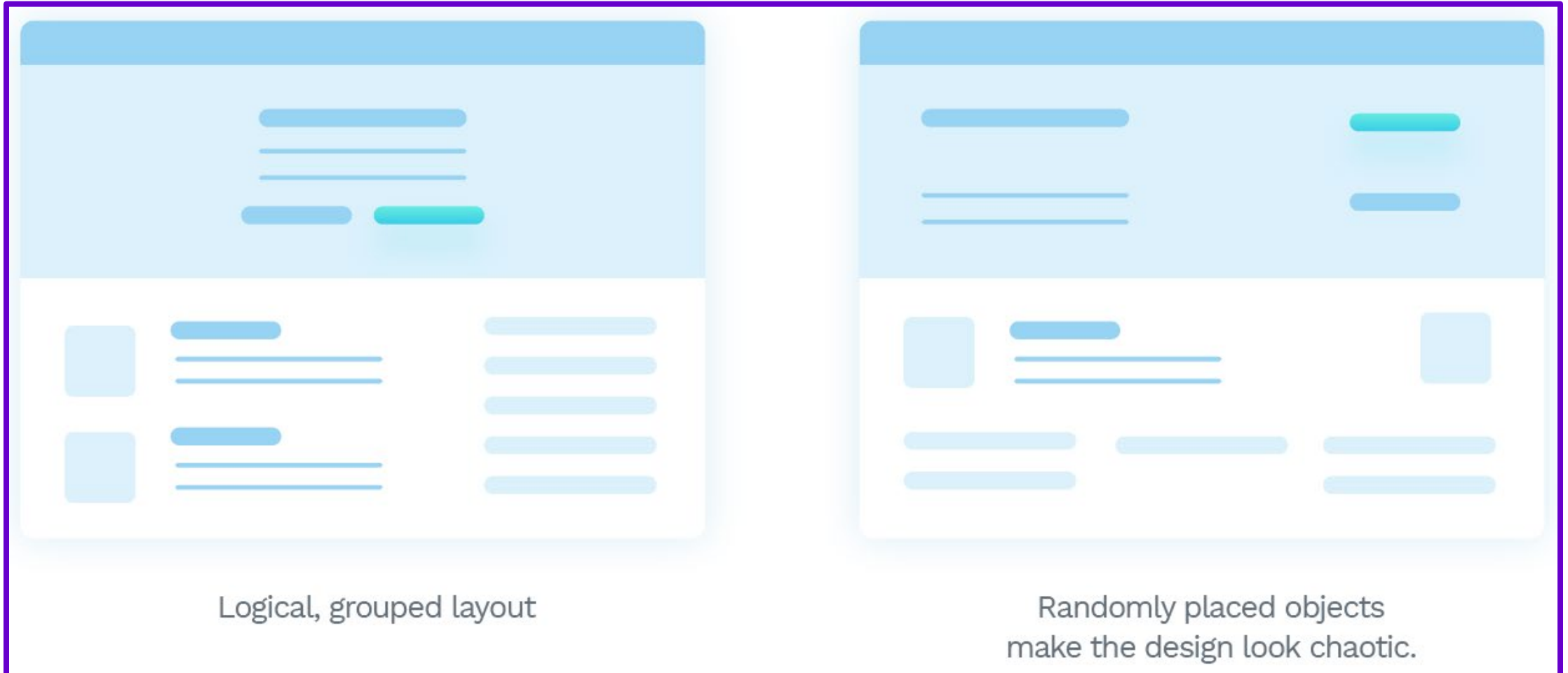
Knowing the rules of perception can make it a lot easier to explain our choices and will help with avoiding many common mistakes.

THE RULE OF PROXIMITY

Objects placed close to each other are automatically understood to be a group.



Chaotic vs Logical Design



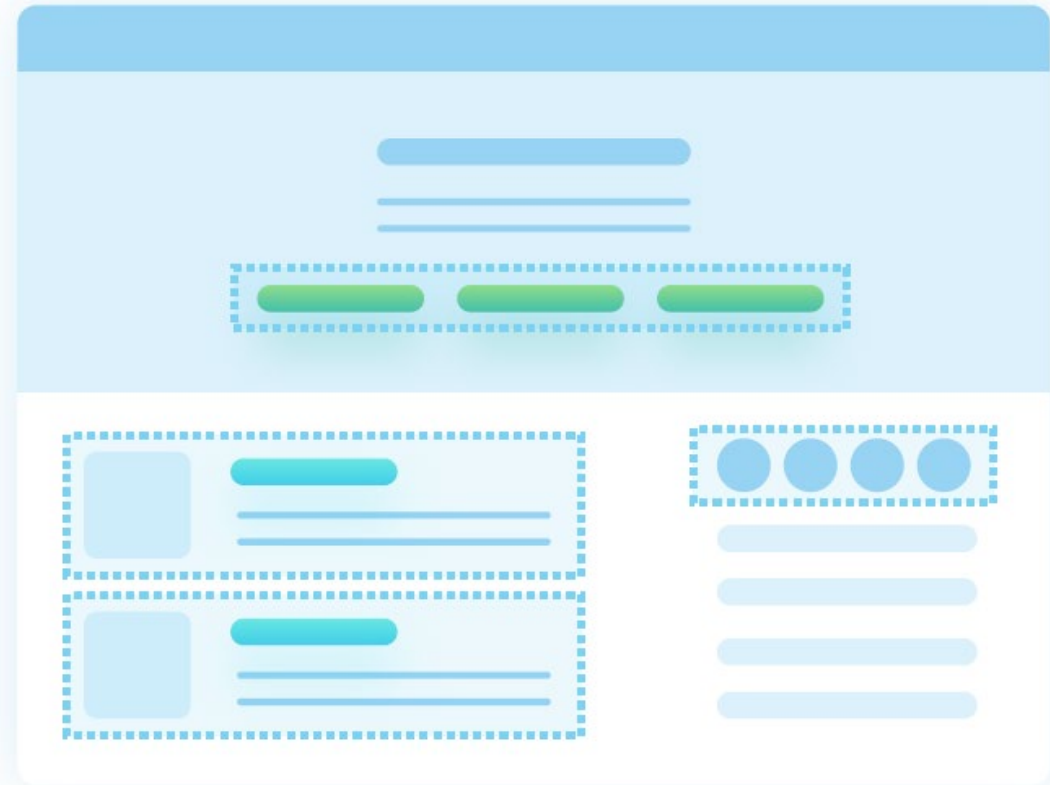
THE RULE OF SIMILARITY

All visually similar objects are considered a part of a group.



But what do we understand as a similarity?

It's all of the defining features like color, shape, size, texture, or position in space.



Our brains groups similar objects together automatically.

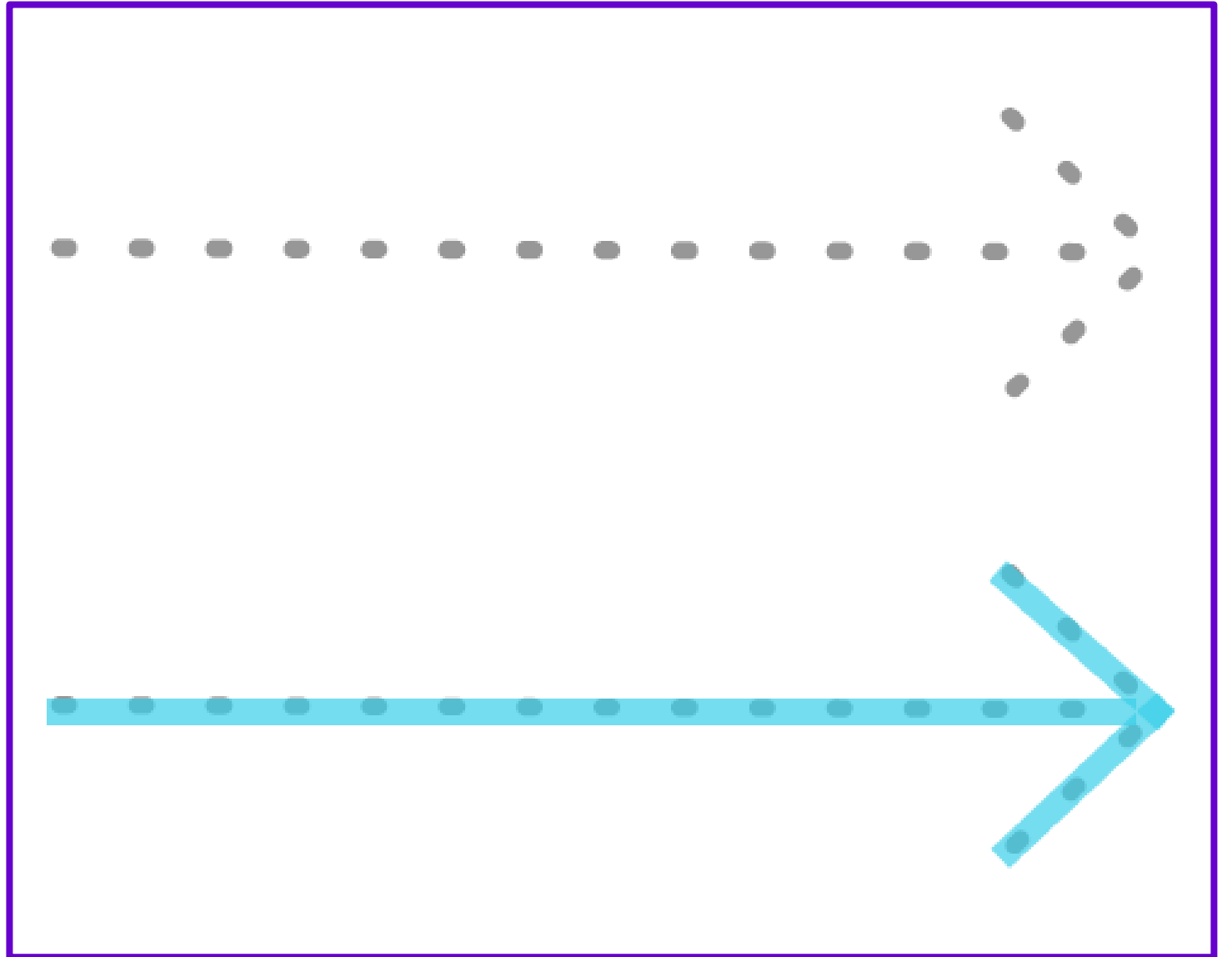
THE RULE OF CLOSURE

A set of objects can naturally form a recognizable shape.



Our brain
fills the blank spaces

We perceive the
result, not as a set
of dots or dashes,
but rather a full
line.



Abstract Shapes

Most icons are purposefully simple not to overload us with information, and our brain fills in the blanks.



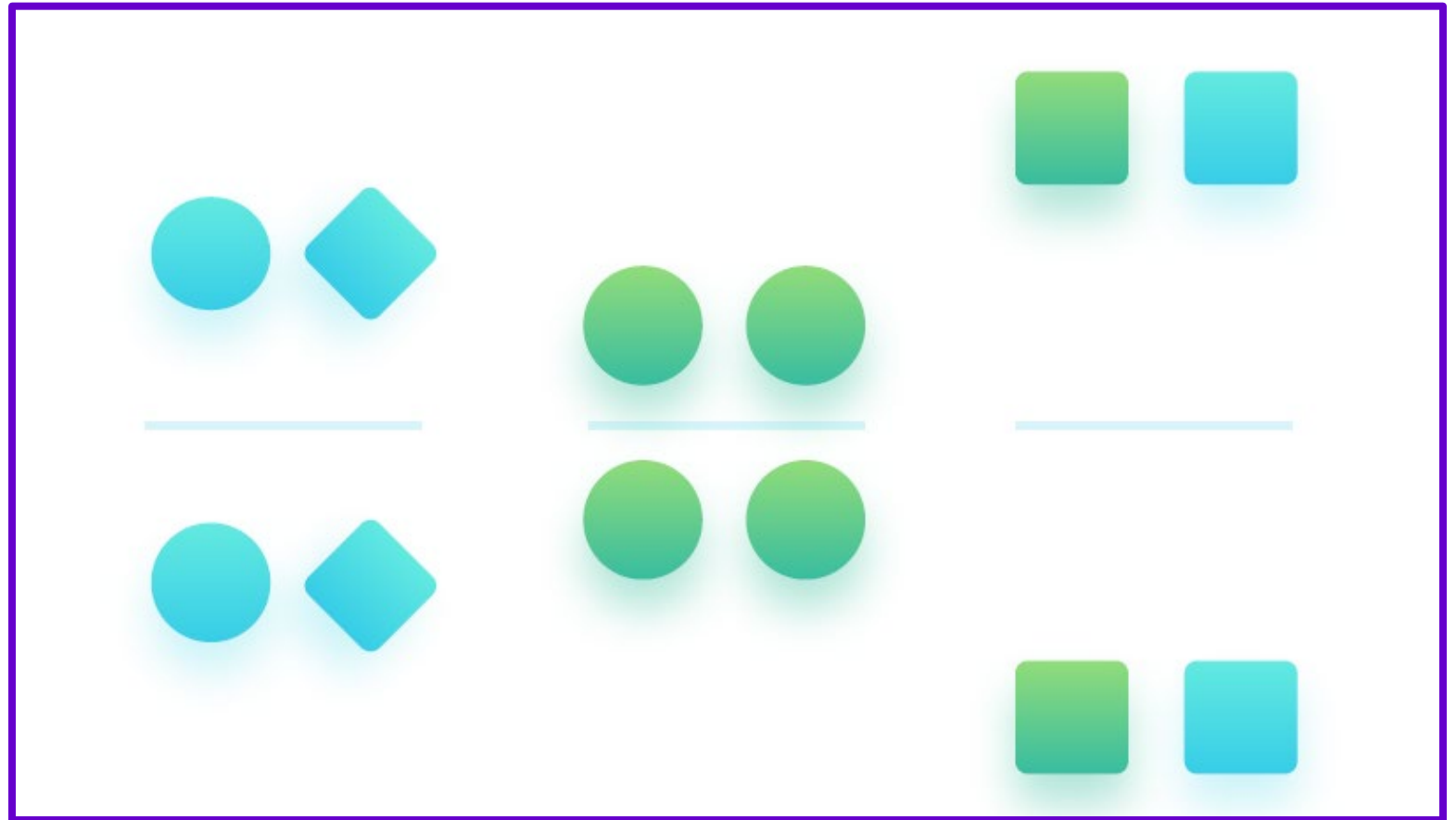
Subconscious
connecting of empty
spaces

Another excellent
example of the
subconscious
connecting of empty
spaces is the WWF
Logo Panda shape or
the IBM logo, where
dashed lines form the
letters

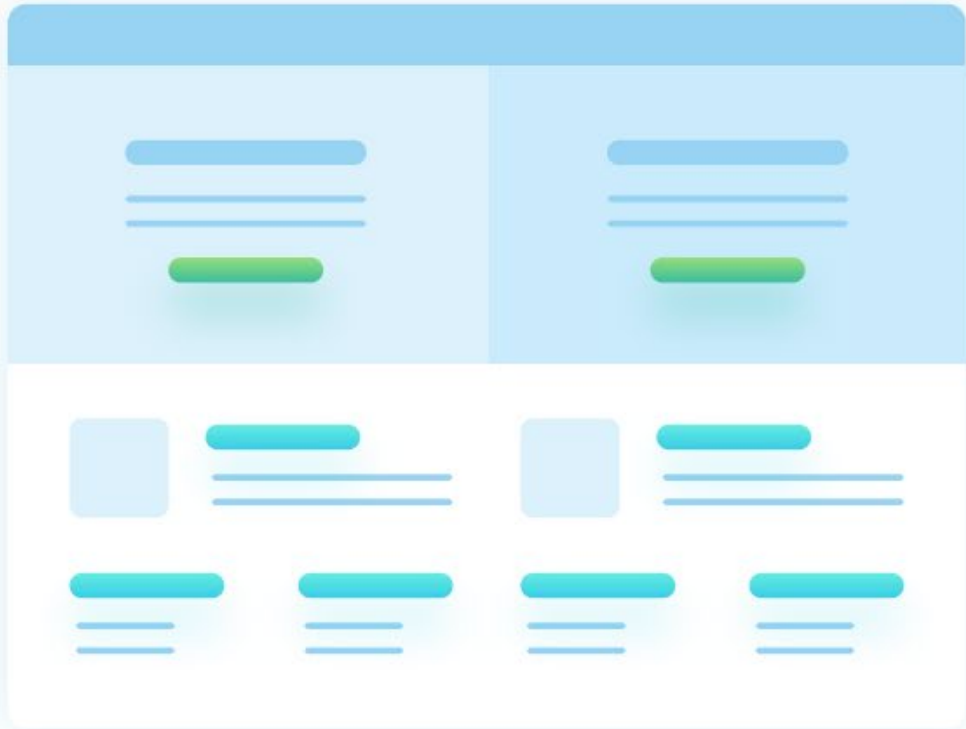


THE RULE OF SYMMETRY

Objects
aligned
symmetrically,
we
understand as
connected



Our brain loves symmetry



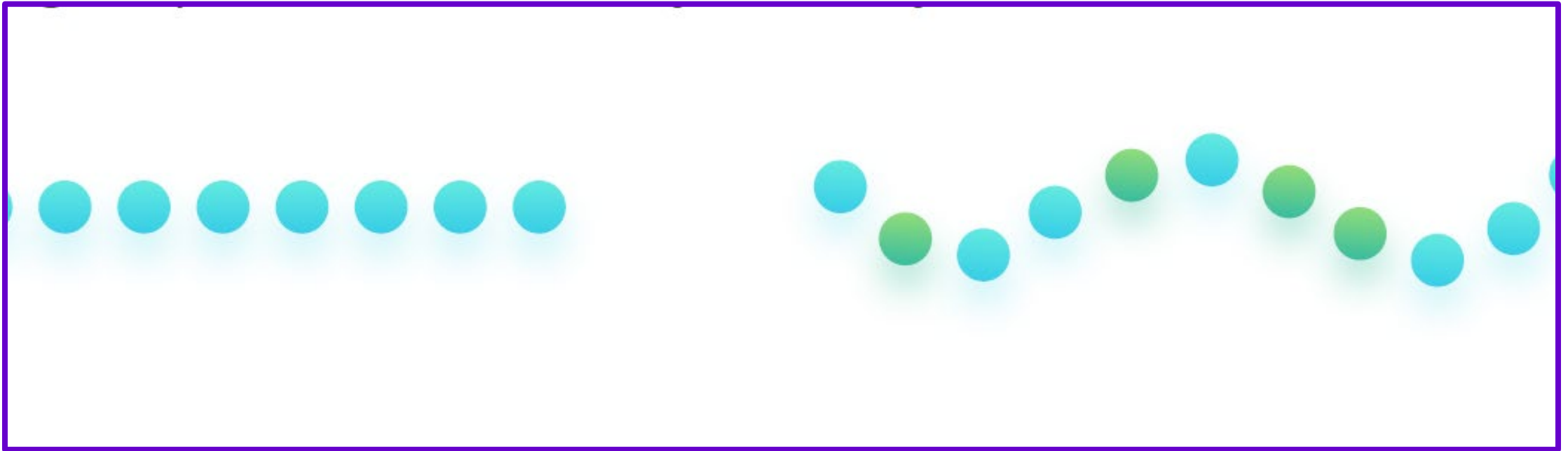
A symmetrical layout like the above will make our design easier to understand and more friendly.



No symmetry causes chaos - we don't know which object to focus on.

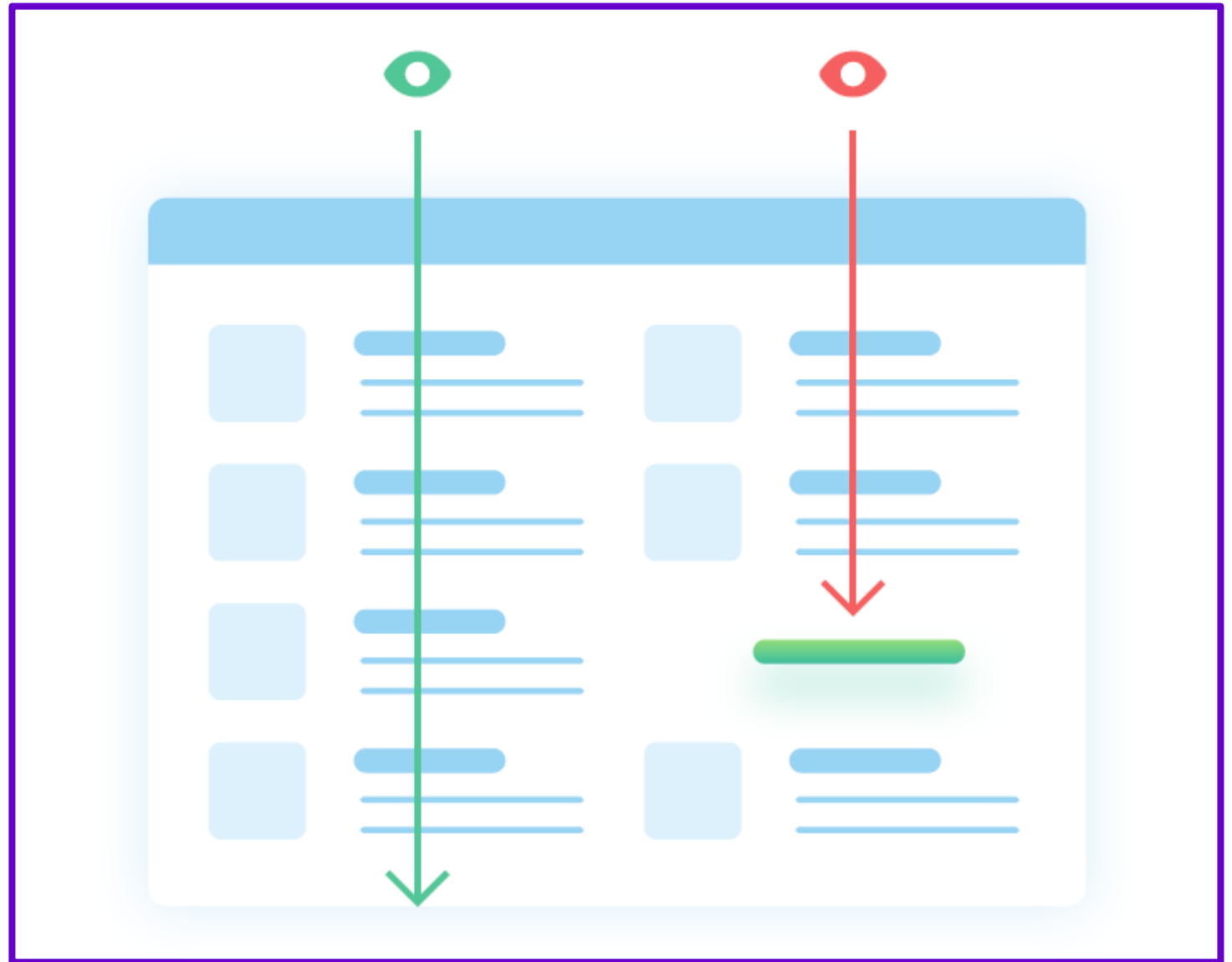
THE RULE OF CONTINUITY

Objects aligned along a continuous line are also naturally grouped.



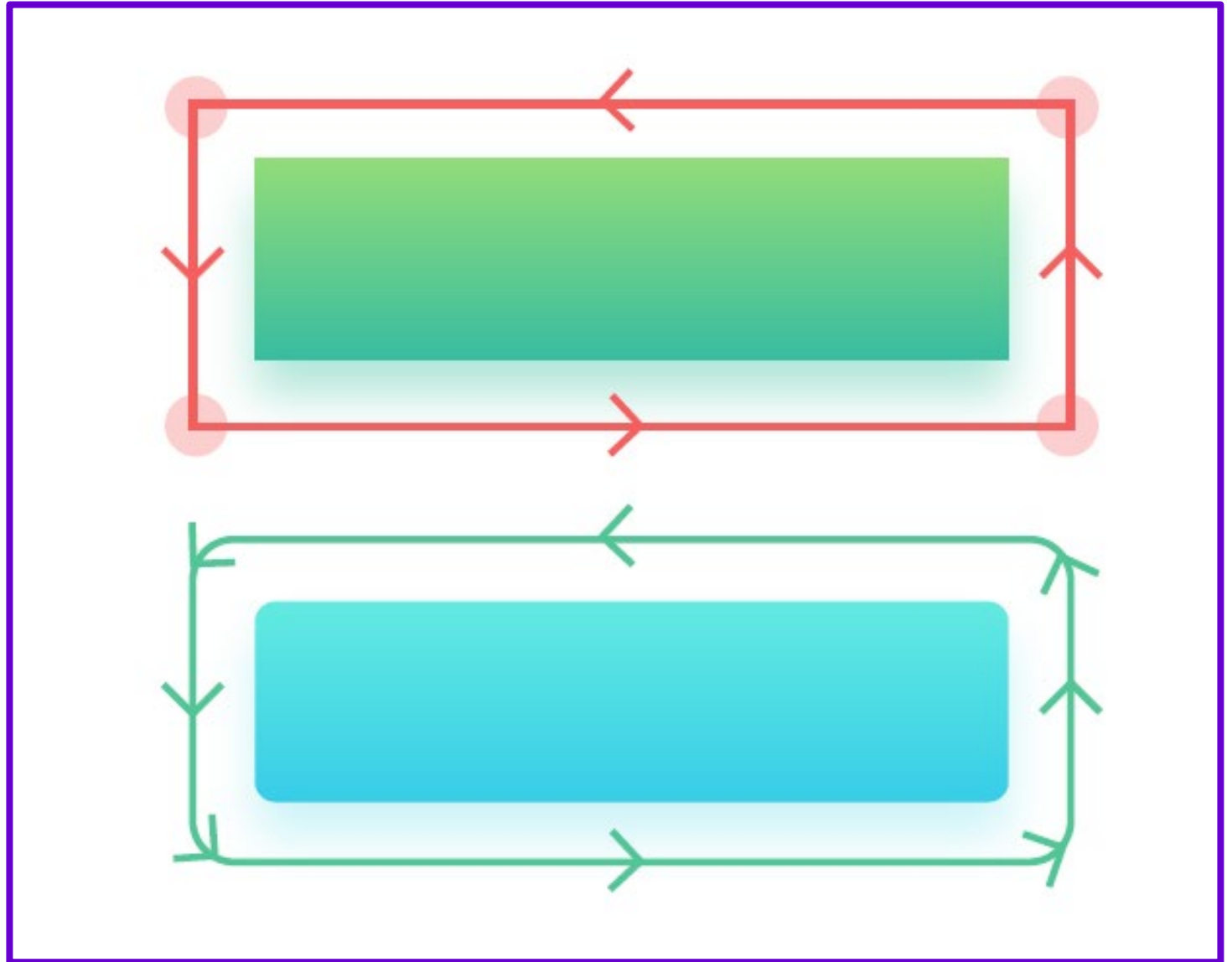
THE RULE OF CONTINUITY

you can see two examples of scanning a page from top to bottom. In the right-side column, a lonely, out of place element breaks the flow.



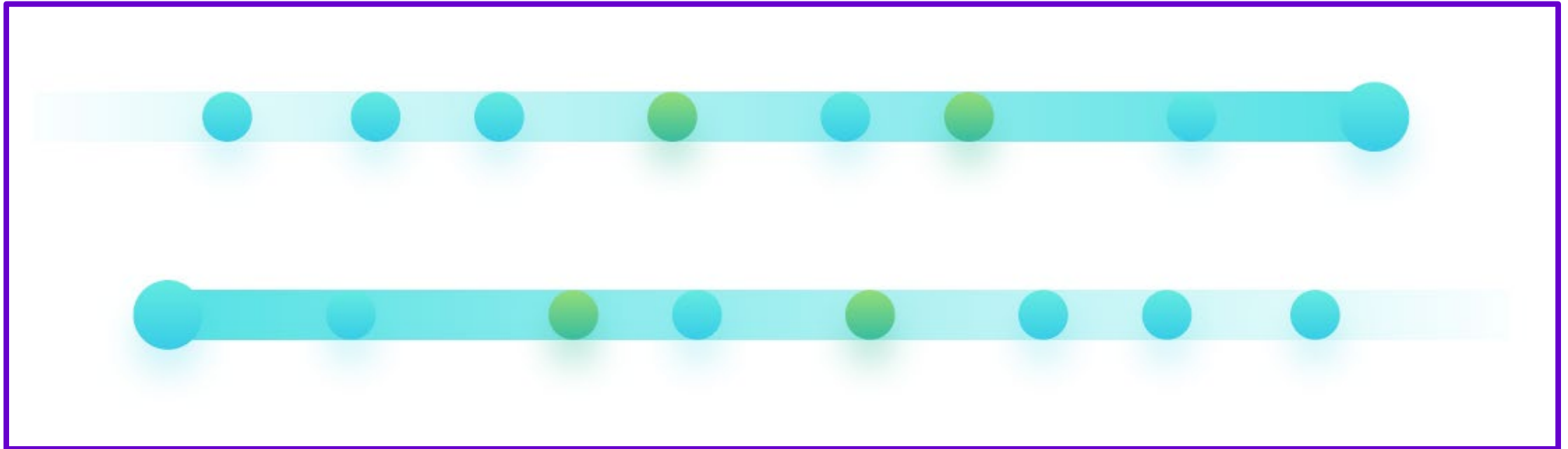
THE RULE OF CONTINUITY

Sharp corners are slightly slower to process, as our eye has to pause and 90-degree turn on each side. A smoother, rounded edge will help our eye go around much faster

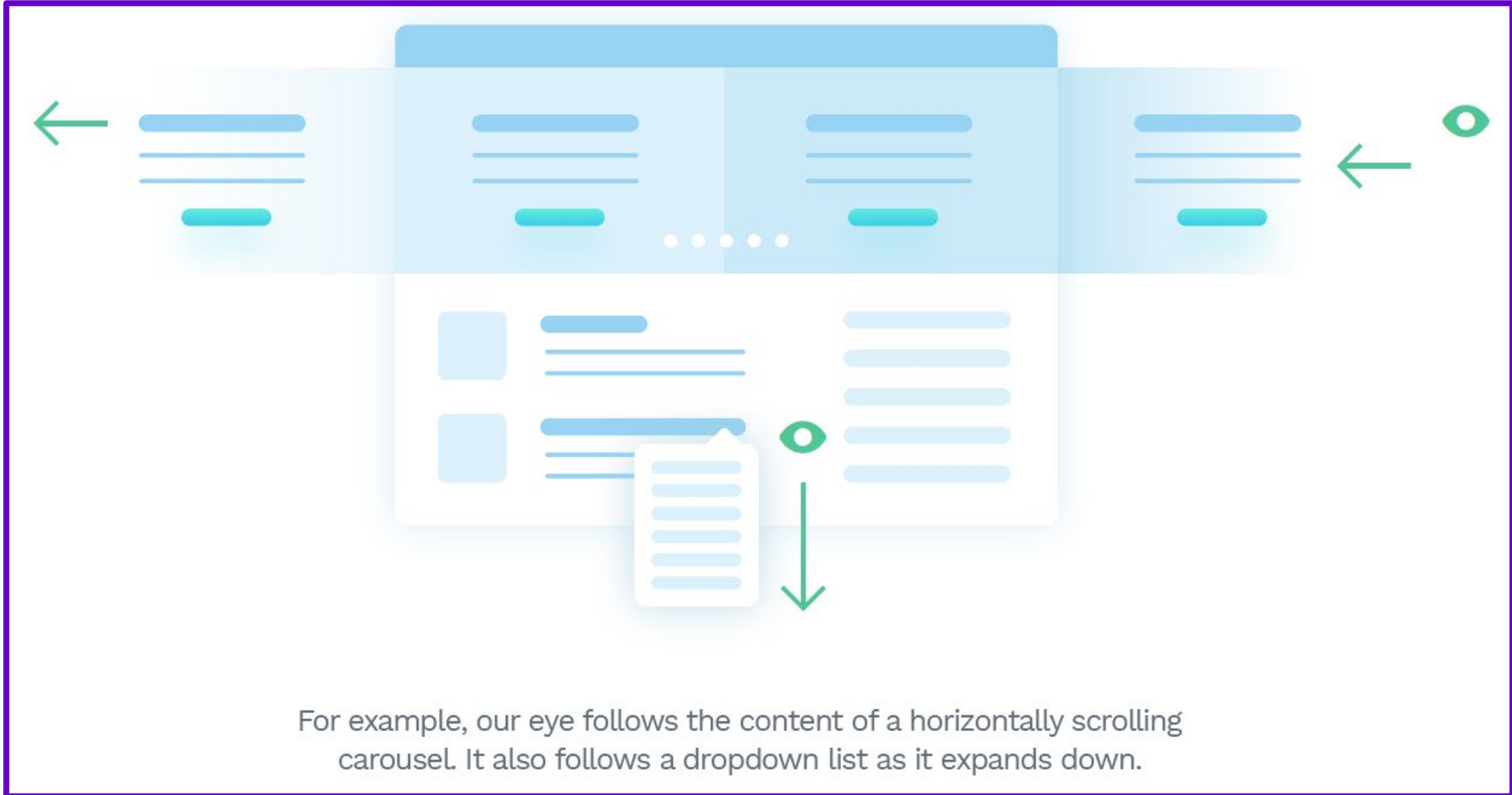


THE COMMON FATE RULE

Objects moving in the same direction, with the same velocity, are considered to be a group.

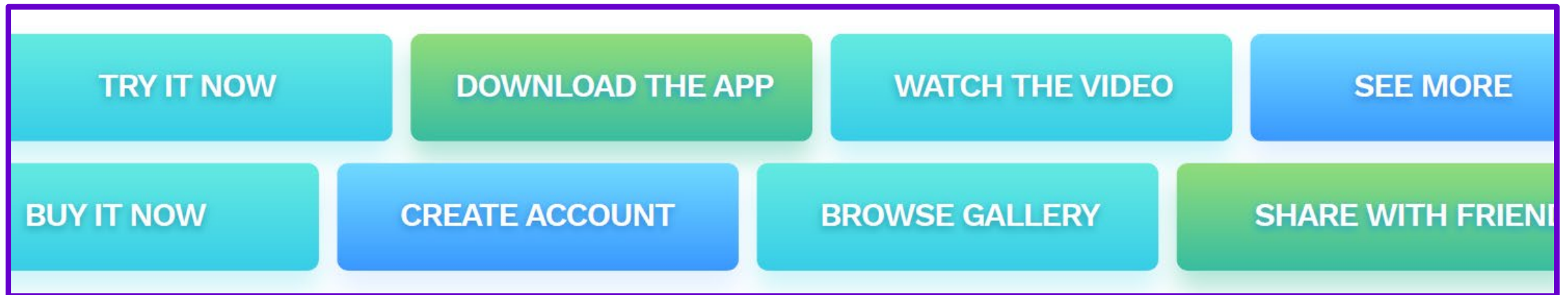


THE COMMON FATE RULE



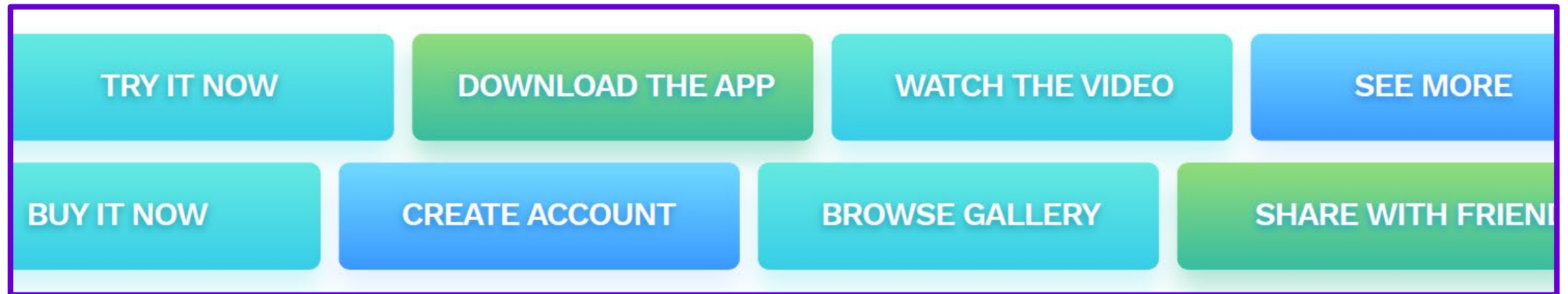
HICK'S LAW

The more options to choose from (and the more complex they are), the harder it is to make a choice



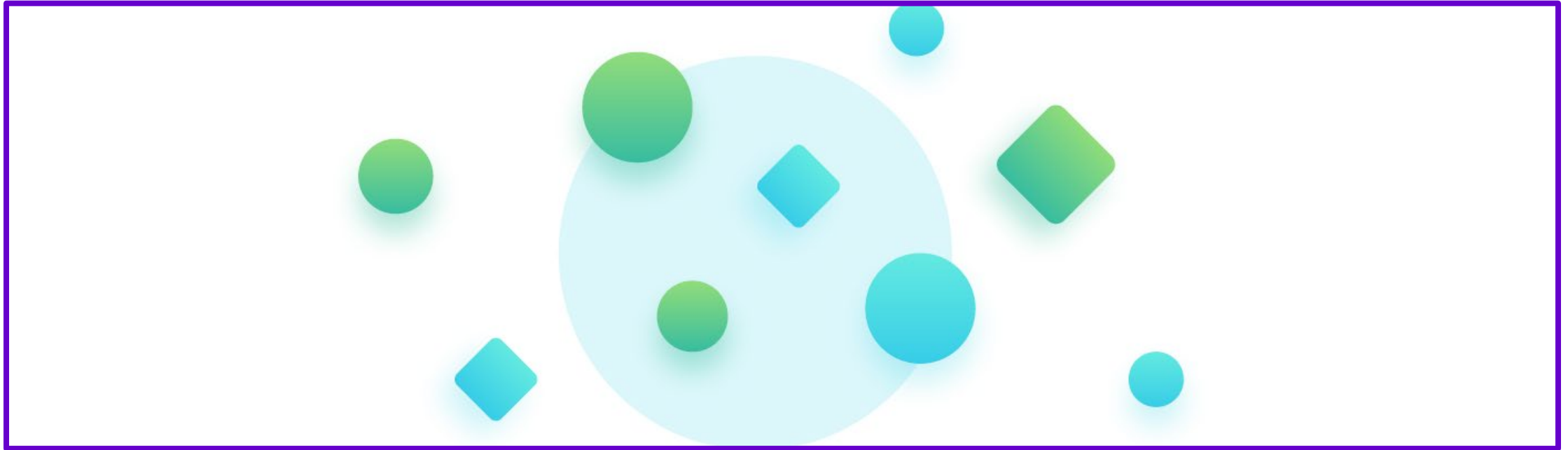
MILLER'S LAW

Our minds can only process around 7 (+/- 2) objects at the same time.



Visual Hirarchy

We instinctively differentiate between an object and a background.



Avoid confusion and breaking hierarchy

You can use contrast, position, and shadows to help the user differentiate the foreground and the background.



THE AESTHETIC USABILITY EFFECT

A visually attractive (high aesthetic) product is often seen as more useful.



THE SERIAL POSITION

We always remember the first and last object in a group best.

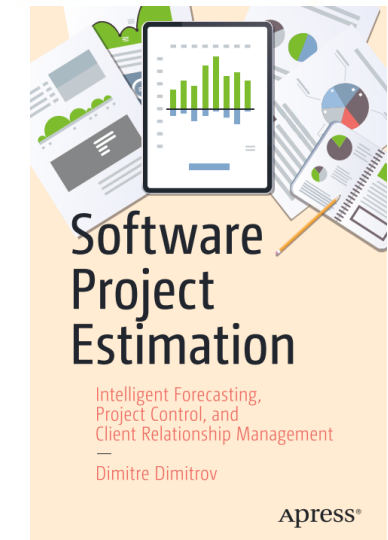
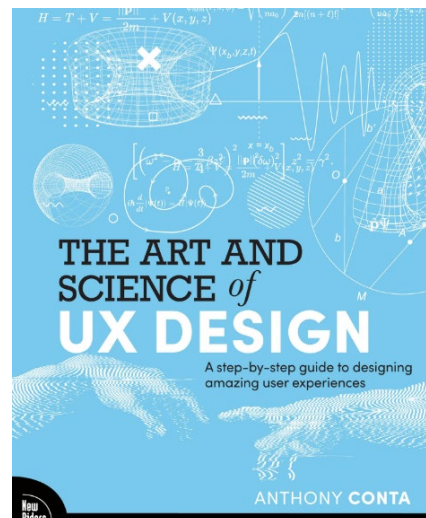
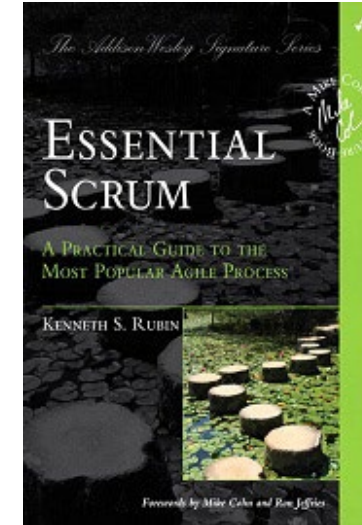
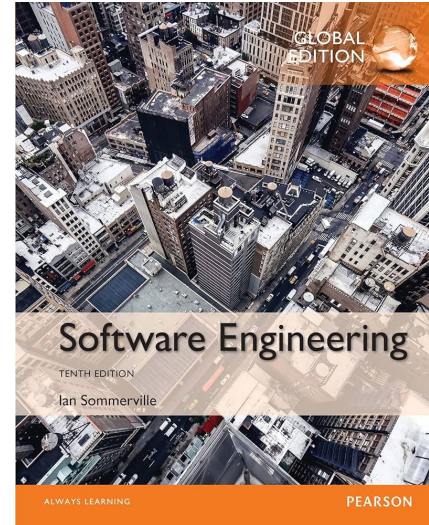
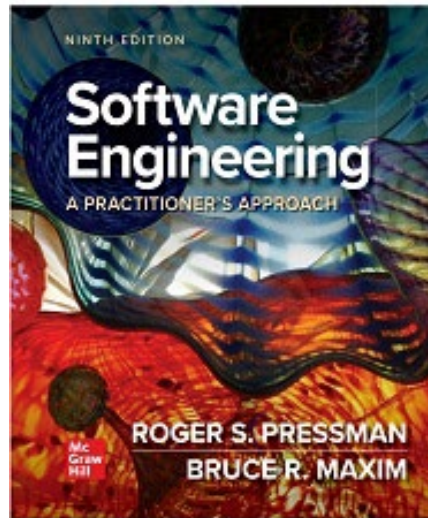


VON RESTROFF'S ISOLATION EFFECT

In a group of similar-looking objects, we always remember the one that doesn't match the rest.



Course References



Course References

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- ***UI Design Styles: Trends and Design Patterns***, Michal Malewicz & Diana Malewice, 2020
- ***What UX Is Really About :Introducing a Mindset for Great Experiences***, Celia Hodent, CRC Press, 2022
- ***Lean UX: Designing Great Products with Agile Teams 3rd Edition***, Jeff Gothelf & Josh Seiden, O'Reilly, 2021
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- [3] Martin, R. C. (2019). Clean Agile, Pearson Education.
- [4] Hall, G. M. (2017). Adaptive Code: Agile coding with design patterns and SOLID principles, Microsoft
- [5] سامانی‌پور، علی. (۲۰۱۸). آموزش اسکرام برای مدیریت چابک فرایند توسعه اپلیکیشن های وب و موبایل،
فرادرس